

# Implement private 5G for teleoperated driving of vehicles in logistics and port operations

Transportation and Logistics × Remote and autonomous operations × Private 5G / LTE · OS082 v1 · refreshed 2026-08-18

| ATTRACTIVENESS                   | RIGHT TO WIN                         | COMPETITION                            | SERVICEABLE MARKET                           | HORIZON                                             | PORTFOLIO              |
|----------------------------------|--------------------------------------|----------------------------------------|----------------------------------------------|-----------------------------------------------------|------------------------|
| <b>68</b><br>is the world moving | <b>66</b><br>can we play, can we win | <b>HIGH</b><br>0 named in the evidence | <b>€2.4bn</b><br>partial evidence · per year | <b>LATER</b><br>research and standards<br>activi... | <b>L0</b><br>8 signals |

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

## The opportunity

This opportunity is to deploy private 5G networks that enable teleoperated driving of vehicles in logistics hubs and port operations, addressing the need for reliable, low-latency remote control. It exists now because teleoperated driving via private 5G has been demonstrated, and private wireless networks are recognized for their determinism and reliability over Wi-Fi, while security challenges are being actively studied. For logistics operators, this means safer, more efficient remote operations in environments where human presence is risky or costly.

## Market opportunity

| Method                                             | Total (TAM)                      | Serviceable (SAM)                | Obtainable (SOM)                | Confidence |
|----------------------------------------------------|----------------------------------|----------------------------------|---------------------------------|------------|
| Bottom-up: enterprises x adoption x contract value | <b>€3.8bn</b><br>€831m – €22.2bn | <b>€2.4bn</b><br>€518m – €13.8bn | <b>€82.8m</b><br>€18.1m – €483m | partial    |
| Observed: tendered contract value, annualised      | <b>€115m</b>                     | <b>€115m</b>                     | <b>€4.0m</b>                    | observed   |

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

### Bottom-up: enterprises x adoption x contract value

| Factor                                                  | Value                                                              | Basis      | Source                                        | As of                  |
|---------------------------------------------------------|--------------------------------------------------------------------|------------|-----------------------------------------------|------------------------|
| Enterprises in Transportation and Logistics (EU27_2020) | 113,199 enterprises                                                | observed   | Eurostat · sbs_sc_ovw                         | 2024                   |
| Enterprises adopting Private 5G / LTE                   | 32.5 % of enterprises (10+ employees)                              | proxy      | Eurostat · isoc_eb_iotn2 · E_IOT1             | 2021                   |
| Annual value of one engagement                          | €792k per enterprise per year                                      | observed   | TED (Tenders Electronic Daily) · ted          | 2026-05-26..2026-08-05 |
| Size-weighted buyer base                                | 14,748 enterprise-equivalents at large-enterprise engagement value | assumption | config/sizing.yaml · size_class_value_weights | size-2026-08-a         |
| Assumed obtainable share of the serviceable market      | 3.5 % of SAM                                                       | assumption | config/sizing.yaml · obtainable_share         | size-2026-08-a         |

**Read this before quoting the number.** Adoption rate is a declared proxy. Internet of Things by NACE Rev. 2 activity series E\_IOT1 is used as a proxy for Private 5G / LTE: No private-network series; industrial IoT use as proxy. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders.

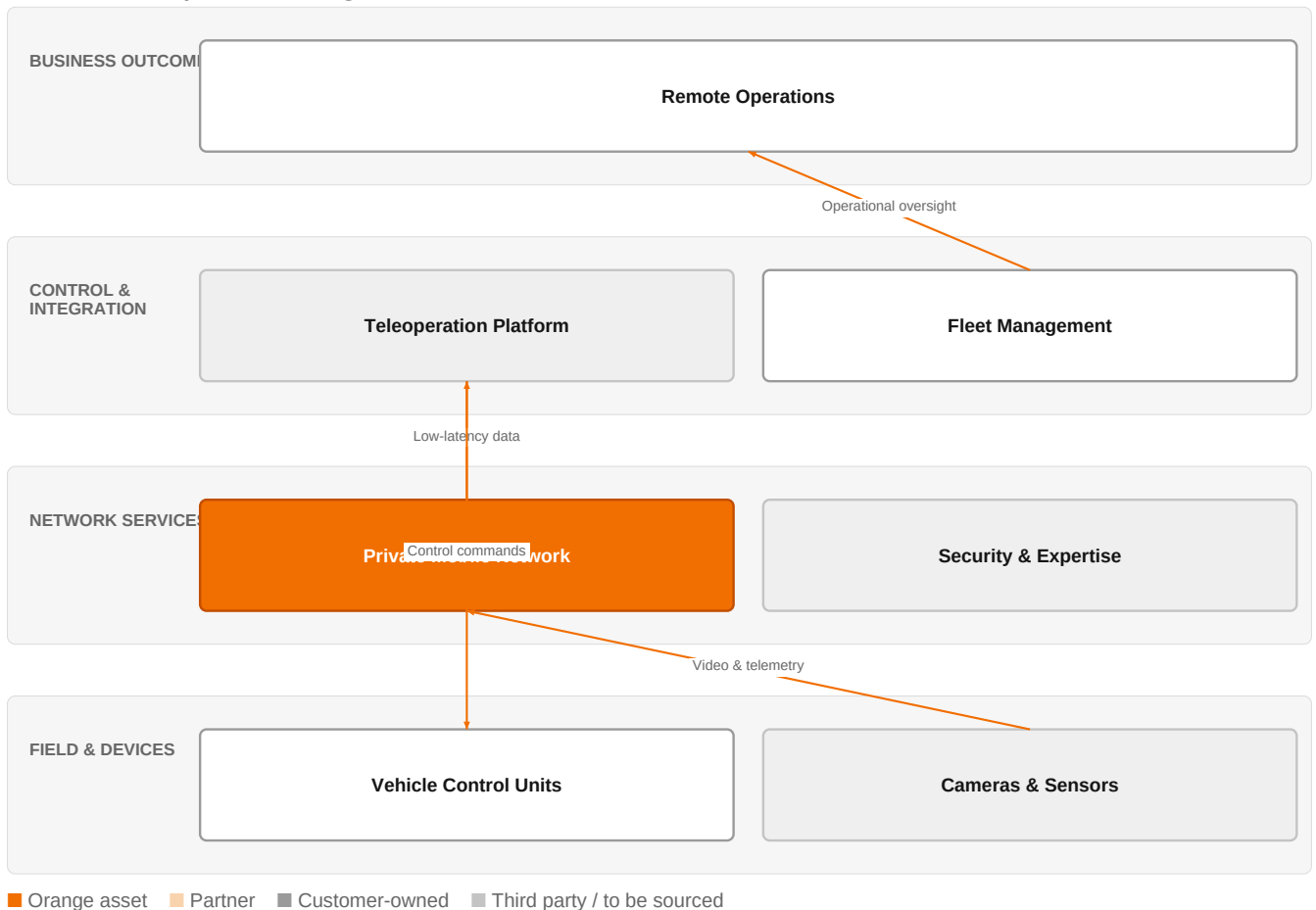
### Observed: tendered contract value, annualised

| Factor                                             | Value          | Basis      | Source                                | As of                  |
|----------------------------------------------------|----------------|------------|---------------------------------------|------------------------|
| Tendered contract value in matching CPV categories | €115m per year | observed   | TED (Tenders Electronic Daily) · ted  | 2026-05-26..2026-08-05 |
| Assumed obtainable share of the serviceable market | 3.5 % of SAM   | assumption | config/sizing.yaml · obtainable_share | size-2026-08-a         |

**Read this before quoting the number.** Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 8 matching notices, matched on vertical via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size.

## The solution, in one picture

### Private 5G Teleoperation for Logistics



This diagram shows how Orange's private mobile network connects sensors and vehicle control units to a teleoperation platform, enabling remote operations with low latency and security.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

## What has changed

Teleoperated driving via a private 5G network has been demonstrated with remotely-driven robots, proving technical feasibility. Private LTE/5G networks are increasingly adopted in Industry 4.0 because they offer greater determinism and reliability than Wi-Fi, which is critical for real-time control. However, these networks present a unique and evolving threat landscape, and security challenges and solutions are being actively researched. The market for private 5G deployments is expanding, with trackers covering thousands of engagements globally, indicating growing enterprise interest.

Sources: SIG-B11748E56A0F 2026 IEEE 12th Interna 2026-06-29; SIG-6FBA261C5571 International Journal 2025-08-08; SIG-4A3CF0B146B7 bing.com 2026-08-12

## Who buys, and why now

The buyer is typically the Chief Operating Officer or Head of Logistics at a port authority or large logistics operator, who is responsible for operational efficiency and safety. The pain is the high cost and risk of manually operating vehicles in hazardous or congested environments, such as container terminals or warehouse yards. The trigger is often a safety incident, labor shortage, or pressure to increase throughput while reducing operational costs, prompting them to explore remote and autonomous operations.

## Why it is hot now — every claim bound to a dated source

- Teleoperated driving via a private 5G network has been demonstrated with remotely-driven robots. [SIG-B11748E56A0F]

## What Orange would deliver, and why it can win

What Orange would deliver

Orange would deliver a private mobile network (L0) deployment, leveraging its position as a Leader in the Gartner Magic Quadrant for 4G and 5G Private Mobile Network Services. The engagement would start with a site survey and network design, then deploy the private 5G/LTE infrastructure, and finally integrate with the customer's vehicle control systems. Orange would bring IoT and Digital experts to ensure seamless integration, and could involve Orange Group researchers for advanced teleoperation challenges. The network would be secured with defense and security experts to address the evolving threat landscape.

#### Why Orange can win

Orange is positioned as a Leader in the Gartner Magic Quadrant for 4G and 5G Private Mobile Network Services, which gives credibility. The capability pool includes IoT, Digital, and Defense and security experts, plus Orange Group researchers, enabling a comprehensive solution. The reference Point One Navigation demonstrates prior experience in precise positioning, which is relevant for teleoperation. However, the assets are thin on specific teleoperation expertise, so Orange must leverage its research arm and partner ecosystem to fill gaps.

#### Named assets behind this — each individually inspectable (LK-08)

| Asset                                                                            | Type             | Link | Why it is linked                                           | Curator     |
|----------------------------------------------------------------------------------|------------------|------|------------------------------------------------------------|-------------|
| Private mobile networks                                                          | offer            | L0   | offer addresses the use case AND provides the technology   | unconfirmed |
| Samsung                                                                          | partner          | L2   | partner provides the required technology                   | unconfirmed |
| Huawei                                                                           | partner          | L2   | partner provides the required technology                   | unconfirmed |
| Gartner Magic Quadrant 2026 — 4G and 5G Private Mobile Network Services (Leader) | analyst position | SUP  | analyst recognition covers this technology or use case     | unconfirmed |
| IoT experts                                                                      | capability pool  | SUP  | capability pool staffs this domain, vertical or technology | unconfirmed |
| Digital experts                                                                  | capability pool  | SUP  | capability pool staffs this domain, vertical or technology | unconfirmed |
| Defense and security experts                                                     | capability pool  | SUP  | capability pool staffs this domain, vertical or technology | unconfirmed |
| Orange Group researchers                                                         | capability pool  | SUP  | capability pool staffs this domain, vertical or technology | unconfirmed |
| Point One Navigation                                                             | reference        | SUP  | published reference in this vertical, different use case   | unconfirmed |

## Competitive landscape

**HIGH** competitive intensity (score 8.0; 0 of 8 listed players appear in this topic's own evidence)

Crowded, with heavyweight incumbents already visible in the evidence. Win on a specific differentiator or do not bid.

| Competitor                   | Category                  | Position here                                                                    | Basis      | Evidence |
|------------------------------|---------------------------|----------------------------------------------------------------------------------|------------|----------|
| BT Business                  | Telecom operator peer     | sells Private 5G / LTE; competes in Connectivity Solutions                       | structural | —        |
| Deutsche Telekom / T-Systems | Telecom operator peer     | sells Private 5G / LTE; competes in Connectivity Solutions, OX: Smart Industries | structural | —        |
| NTT DATA                     | Global systems integrator | sells Private 5G / LTE; competes in Connectivity Solutions, OX: Smart Industries | structural | —        |
| Proximus                     | Telecom operator peer     | sells Private 5G / LTE; competes in Connectivity Solutions                       | structural | —        |
| Swisscom                     | Telecom operator peer     | sells Private 5G / LTE; competes in Connectivity Solutions                       | structural | —        |
| Telefonica Tech              | Telecom operator peer     | sells Private 5G / LTE; competes in Connectivity Solutions                       | structural | —        |
| Telia Company                | Telecom operator peer     | sells Private 5G / LTE; competes in Connectivity Solutions, OX: Smart Industries | structural | —        |
| Verizon Business             | Telecom operator peer     | sells Private 5G / LTE; competes in Connectivity Solutions                       | structural | —        |

**evidenced** means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

## Qualifying questions for the first meeting

- 1 What specific vehicles or equipment in your operations are candidates for teleoperation, and what is the current manual process?
- 2 Have you experienced any safety incidents or operational bottlenecks that remote operation could address?
- 3 What is your current wireless infrastructure (Wi-Fi, public cellular) and why is it insufficient for remote control?
- 4 Do you have any existing partnerships or pilots with technology providers for autonomous or remote operations?
- 5 What are your latency and reliability requirements for control commands and video feeds?
- 6 Are there regulatory or labor constraints that make remote operation attractive?

## Objections you will meet

| They will say                                                                     | You can answer                                                                                                                                                                                                                                                |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| We already have Wi-Fi, why do we need private 5G?                                 | Wi-Fi can work for some applications, but private 5G offers greater determinism and reliability, which is critical for teleoperation where latency and packet loss can be dangerous. Our private mobile network is designed for industrial-grade performance. |
| Teleoperation is still experimental; we are not convinced it is production-ready. | It is true that teleoperation over private 5G is early, but it has been demonstrated successfully with robots. We can start with a pilot to prove the business case in your environment, leveraging our research and integration expertise.                   |
| Security is a major concern for us.                                               | Private wireless networks do present a unique threat landscape, but we have defense and security experts who can design a secure architecture. We address security from day one, not as an afterthought.                                                      |

## Risks and unknowns

The main risk is that teleoperation over private 5G is still emerging; while demonstrated, it is not yet a standard commercial offering. Security concerns around private wireless networks could slow adoption. The business case depends on the customer's specific operational pain points and regulatory environment. Unknowns include the actual latency requirements for different vehicle types and the integration complexity with existing fleet management systems.

## Next action, by role (FR-17)

| Role                   | Do this next                                                                                                                                                                                                                                                                                     |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Presales / Proposal    | Develop a bid brief for a logistics or port operator, differentiating our private 5G offering by citing the Gartner Magic Quadrant 2026 Leader position and the successful demonstration with Point One Navigation, and include Samsung and Huawei as technology partners.                       |
| Sales                  | In a meeting with a logistics or port operator's operations director, say: 'We have demonstrated teleoperated driving over private 5G with remotely-driven robots, and our private mobile networks can support your remote and autonomous operations.'                                           |
| Strategist / Innovator | Commission a deep-dive brief on the technical and commercial viability of private 5G for teleoperated driving in logistics and port operations, leveraging Orange Group researchers and IoT experts, and assess the competitive landscape using the Gartner Magic Quadrant 2026 Leader position. |

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

| Signal           | Date       | Publisher                    | T | Title and link                                                                                     |
|------------------|------------|------------------------------|---|----------------------------------------------------------------------------------------------------|
| SIG-5EF8BAA50BF5 | 2026-07-14 | Iconic Research and Engin... | 1 | 5G Core SA Enablement for Private Industrial Networks: Opportunities for Oil, Gas, Mining, Logi... |
| SIG-B11748E56A0F | 2026-06-29 | 2026 IEEE 12th Internatio... | 1 | Teleoperated Driving via a Private 5G Network: Demonstration with Remotely-Driven Robots           |
| SIG-6FBA261C5571 | 2025-08-08 | International Journal on ... | 1 | Security Challenges and Solutions in Private Wireless (LTE/5G) Network Implementations             |
| SIG-CC7D9567F38D | 2025-07-31 | International Journal for... | 1 | Private 5G Network: Security, Challenges, and Comparison with Wi-Fi                                |
| SIG-9E3ACF1CE0DC | 2025-07-16 | Emitor: Jurnal Teknik Ele... | 1 | Implementation of 5G Standalone Private Network for UHD Video Streaming                            |
| SIG-C0910B893CD2 | 2024-06-01 | cordis.europa.eu             | 1 | Preparing the ground for AUTonomous Multimodal SUPply Chains                                       |
| SIG-4A3CF0B146B7 | 2026-08-12 | bing.com                     | 2 | Private 5G Network Deployment Tracker & Forecasts 2026-2030   Covers 9,300 Engagements Across 1... |
| SIG-CDD2DA807ACB | 2026-08-11 | arxiv.org                    | 3 | A Neural Network Based Teleoperation for Remote Controlled Vehicles                                |

How this brief was made

**Computed, not written.** Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

**Written, then checked.** The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: competitive\_landscape (no claim survived evidence binding); diagram (1 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

| What                        | Version                     | When                      |
|-----------------------------|-----------------------------|---------------------------|
| Scores — weight set         | w-2026-08-a                 | 2026-08-18T19:08:26+00:00 |
| Market sizing — assumptions | size-2026-08-a              | 2026-08-18T21:06:14+00:00 |
| Competitor register         | competitors-2026-08-a       | 2026-08-18T21:06:17+00:00 |
| Narrative — prompt / model  | describe-v1 / deepseek-chat | 2026-08-19T04:56:42+00:00 |
| Pipeline                    | 0.1.0                       | 2026-08-18                |

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.